

GUJARAT TECHNOLOGICAL UNIVERSITY
BE –SEMESTER 1&2(NEW SYLLABUS)EXAMINATION- WINTER 2018

Subject Code:3110003**Date: 09-01-2019****Subject Name: PROGRAMMING FOR PROBLEM SOLVING****Time: 10:30 am to 01:00 pm****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

- Q.1** (a) Define: algorithm, flowchart, compiler **03**
 (b) Why null value is used in string? Justify your answer with example **04**
 (c) List all operators used in C and explain any three with example. **07**
- Q.2** (a) What is UDF? Describe advantages of UDF. **03**
 (b) Draw all symbols used in flowchart and draw flowchart to find factorial number. **04**
 (c) Explain while loop with example. **07**
- OR**
- (c) Explain for loop with example. **07**
- Q.3** (a) Describe local and global variable with example. **03**
 (b) Explain break and continue statement with example. **04**
 (c) Write a C program to input an integer number and check last digit of number is even or odd. **07**
- OR**
- Q.3** (a) Define general form of 1) do while loop 2) Nested if 3) goto **03**
 (b) Describe precedence and associativity of operators with example. **04**
 (c) Explain switch case statement with example to read number between 1 to 7 and print relatively day Sunday to Saturday. **07**
- Q.4** (a) What is pointer? Explain with example to print the address of variable using pointer. **03**
 (b) Explain fopen() and its mode with example to write a string into file **04**
 (c) Define a structure data type called time_struct containing three member's integer hours, minutes, second. Develop a program that would assign values to individual member and display the time in following format : HH:MM:SS **07**
- OR**
- Q.4** (a) Describe array with example. **03**
 (b) What is string? Explain with example to find a character from string. **04**
 (c) What is structure? How to access the elements of structure? How to calculate size of structure? Explain with example. **07**
- Q.5** (a) Define recursion. List the advantages of recursion. **03**
 (b) Classification of User defined function's components (elements) **04**
 (c) Create a function to check number is prime or not. If number is prime then function return value 1 otherwise return 0. **07**
- OR**
- Q.5** (a) What do you mean by type conversion? Give example. **03**
 (b) Discuss parameter passing technique used in C with example. **04**
 (c) Create a function to swap the values of two variables. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I &II (NEW) EXAMINATION – SUMMER-2019****Subject Code: 3110003****Date: 10/06/2019****Subject Name: Programming for Problem Solving****Time: 10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Write flowchart or algorithm to find area of a triangle.	3
	(b) Write a program to reverse a given number.	4
	(c) Explain various looping control structures with suitable example.	7
Q.2	(a) What are header files? Name at least 3 with its usage.	3
	(b) Write a program to find $1+1/2+1/3+1/4+....+1/n$.	4
	(c) What is a string? Explain at least 4 built-in string functions with example.	7
	OR	
	(c) What is an array? Explain one dimensional and two dimensional array declarations and initialization with suitable example.	7
Q.3	(a) What is formatted output? Using printf() statement explain it.	3
	(b) Write a program to check whether entered character is vowel or not?	4
	(c) Write a program to print all Armstrong numbers in a given range. Armstrong number is equal to sum of cubes of its individual digits. For example $153 = 1^3 + 5^3 + 3^3$. So, 153 is Armstrong number.	7
	OR	
Q.3	(a) Why it is necessary to give the size of an array in array declaration?	3
	(b) Explain break and continue with suitable example.	4
	(c) Write a program to display transpose of given 3*3 matrix.	7
Q.4	(a) What is pointer? Which arithmetic operations are not valid on pointers?	3
	(b) Explain array of pointers with suitable example.	4
	(c) Write a program to calculate nCr using user defined function. $nCr = n! / (r! * (n-r)!)$	7
	OR	
Q.4	(a) What is pointer? Which arithmetic operations are valid on pointers?	3
	(b) What is pointer to pointer? Write suitable example to demonstrate the concept.	4
	(c) What is recursive function? Explain with suitable example.	7
Q.5	(a) What care must be taken while writing a program with recursive function?	3
	(b) Explain how structure variable is initialized with suitable example.	4
	(c) What are command line arguments? Explain with suitable example.	7
	OR	
Q.5	(a) In user defined function, what is actual argument and formal argument?	3
	(b) Explain with suitable example structure variable and pointer to structure variable.	4
	(c) What is dynamic memory allocation? Explain important functions associated with it.	7

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– I & II (NEW) EXAMINATION – WINTER 2019****Subject Code: 3110003****Date: 07/01/2020****Subject Name: Programming for Problem Solving****Time: 10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		MARKS
Q.1	(a) List out types of software with Examples.	03
	(b) Distinguish the data types provided by C programming language.	04
	(c) Implement a C Program to convert temperature from Fahrenheit to Celsius and vice versa.	07
Q.2	(a) Define break and continue statement with example.	03
	(b) Categorize the major components of computer system and give their function.	04
	(c) List all symbols used in flowchart and draw flowchart to find factorial number.	07
	OR	
	(c) Construct 'C' program to print the following pattern using loop statement.	07
	1	
	2 2	
	3 3 3	
	4 4 4 4	
	5 5 5 5 5	
Q.3	(a) Explain different types of constants.	03
	(b) Define algorithm and explain different symbols used in flowchart.	04
	(c) Demonstrate a C program to input an integer number and check last digit of number is even or odd.	07
	OR	
Q.3	(a) Explain getch(), getchar(), gets().	03
	(b) List out the operators used in C language and explain any three with example	04
	(c) Write a program to find sum of first N odd numbers. Ex. 1+3+5+7+.....+N	07
Q.4	(a) Show the important of stdio.h header file.	03
	(b) Describe file management. And List the various file management functions.	04
	(c) Build a function to check number is prime or not. If number is prime then function return value 1 otherwise return 0.	07
	OR	
Q.4	(a) Distinguish between Structure and Union.	03
	(b) Develop an algorithm to print first N Fibonacci numbers.	04

- | | | | |
|------------|-----|--|-----------|
| | (c) | Write a C program to read 10 numbers from user and store them in an array. Display Sum, Minimum and Average of the numbers | 07 |
| Q.5 | (a) | Write a program to illustrate the use of fputc () and fputs() | 03 |
| | (b) | Categorize User defined function's components (elements). | 04 |
| | (c) | Explain the function definition, function prototype and function call with relative example. | 07 |
| | | OR | |
| Q.5 | (a) | List the advantages of recursion. | 03 |
| | (b) | Explain for loop with example. | 04 |
| | (c) | Explain call by value (pass by value) and call by reference (pass by reference) with examples in brief. | 07 |

Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE – SEMESTER 1&2 EXAMINATION – SUMMER 2020

Subject Code: 3110003

Date: 05/11/2020

Subject Name: Programming For Problem Solving

Time: 10:30 AM TO 01:00 PM

Total Marks: 70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Explain different symbols used in flowchart.	03
	(b) Write algorithm or flowchart for finding maximum out of three numbers entered by user.	04
	(c) Explain different categories of operators available in C programming.	07
Q.2	(a) Explain entry controlled and exit controlled loop with proper syntax.	03
	(b) Explain 1) nested if-else statement 2) break statement.	04
	(c) Write a program to store 10 elements in array given by user and to find maximum out of those 10 elements.	07
	OR	
	(c) Write a program to find a character from the string, string and character to be searched both will be given by user.	07
Q.3	(a) Show 1D array declaration, initialization and iteration.	03
	(b) Explain type conversion and type casting with example.	04
	(c) Write a C program to read numbers 1 to 7 and print relatively day Sunday to Saturday using switch statement.	07
	OR	
Q.3	(a) List any three header file with its usage.	03
	(b) Explain four basic data types in C and provision to extend range for appropriate data types.	04
	(c) Write a C program to find factorial of a given number.	07
Q.4	(a) Write a program to find area of triangle.	03
	(b) Explain problem solving using recursion with any example.	04
	(c) What is user defined function? Explain pass by value and pass by reference in functions with example.	07
	OR	
Q.4	(a) Show 2D array declaration, initialization and iteration.	03
	(b) Explain any four inbuilt string functions.	04
	(c) Write a function in the program to return 1 if number passed is prime otherwise return 0.	07
Q.5	(a) Explain various file management function in C.	03
	(b) Show structure declaration and initialization with proper syntax.	04
	(c) Write a C program to find $1+1/2!+1/3!+1/4!+.....+1/n!$.	07

OR

- Q.5** (a) What is pointer? Explain with example to store and print the address of variable using pointer **03**
- (b) Show the usage of malloc() and calloc() with syntax. **04**
- (c) Define a structure “personal” that would contain person name, date of joining and salary. Using this structure read information of 5 people and print the same on screen. Also display sum of salary of all 5 people. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-1/2 EXAMINATION – WINTER 2021****Subject Code:3110003****Date:24/03/2022****Subject Name:Programming for Problem Solving****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define following terms: 1) Application Software 2) System Software 3) Algorithm 4) Flowchart.	04
	(b) Draw the flow chart to find the largest of the given three numbers – A ,B and C	03
	(c) Explain different type of operators used in c language with their precedence and associativity.	07
Q.2	(a) Discuss use of break and continue statement in C with example.	03
	(b) Compare and contrast while and do while loop with example.	04
	(c) Write a C program to print following pattern. 1 2 2 3 3 3	07
	OR	
	(c) Write a C program to print following pattern. 1 2 3 4 5 6	07
Q.3	(a) Write a program to check whether entered character is vowel or not?	03
	(b) Explain getch(), getchar(), gets(), puts() .	04
	(c) Develop a menu-based program to perform addition, multiplication, subtraction and division using user-defined function.	07
	OR	
Q.3	(a) Write an algorithm for finding odd and even number from given two numbers.	03
	(b) Write a program to check whether entered number is prime or not with the help of user-defined function check-prime().	04
	(c) Write a program to find out the largest of an array.	07
Q.4	(a) What is structure? Explain with example how to declare a structure and how to initialize it.	03
	(b) Explain following string manipulation function. strcat(), strcpy() ,strcmp() and strlen()	04
	(c) Write a program in c for multiply two matrices A and B of dimensions pXq and qXr respectively and store the result in third matrix C.	07

OR

Q.4	(a)	Demonstrate declaration and initialization of two dimensional array with suitable example.	03
	(b)	Explain nested if else ladder with suitable example.	04
	(c)	Write a program in c using structure to enter rollno, marks of the three subject for 3 student and find total obtained by each student	07
Q.5	(a)	What do you mean by recursive function? What care must be taken while writing a program with recursive function?	03
	(b)	Explain fopen() and its mode with example.	04
	(c)	Describe different categories of user-defined functions.	07
OR			
Q.5	(a)	What is pointer? Explain how pointers are declared and initialized. .	03
	(b)	Compare malloc() and calloc() functions for dynamic memory allocation.	04
	(c)	Develop a program in C to check the entered number is prime or not by creating a user-defined function named check_prime().	07

Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II(NEW)EXAMINATION – SUMMER 2022

Subject Code:3110003

Date:06-08-2022

Subject Name:Programming for Problem Solving

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

Q.1	Mark
(a) Answer the following questions	03
1. Define: Compiler	
2. Justify, 2 and '2' both are not same in C language.	
3. What is the role of sizeof operator in C language?	
(b) Answer the following questions	04
1. Ternary operator can be nested. (True/False)	
2. What do you mean by enumerated data type in C language?	
3. What is the difference between <code>char *p</code> and <code>char p[]</code> ?	
4. What do you mean by function prototype?	
(b) Answer the following questions	07
1. Define Algorithm.	
2. What do you mean by recursion?	
3. What do you mean by program and program control?	
4. What is a pointer?	
5. What is the role of <code>getc()</code> and <code>getw()</code> file functions?	
6. Programming needs logic building. Justify	
7. Define Interpreter.	
Q.2 (a) Write the outputs of the following expression:	03
i) $50 \% 2 / 3 + 2$	
ii) $21 / (\text{int}) 2.5 + 3$	
iii) $(1 > 2) \parallel (2 < 3) \&\& 5 < 1$	
(b) Write a program to print Fibonacci series.e.g.0, 1, 1, 2, 3, 5, 8, 13.	04
(c) Explain the working of various bit-wise operators with the example.	07
OR	
(c) List out all various string functions and describe them with syntax and example.	07
Q.3 (a) Explain ternary (?:) operator in detail with the example.	03
(b) Discuss the need of break and continue statements with example.	04
(c) Design a flowchart for checking whether a given number is palindrome or not.	07
OR	
Q.3 (a) What is meant by array of pointers? Explain it with example.	03

- (b) Write a program to print the triangle shown below. 04
- ```

A
A B
A B C
A B C D

```
- (c) Draw block diagram of computer system and explain the functions of each component in detail. 07
- Q.4** (a) Explain type casting with example. 03
- (b) Compare and contrast goto statement and switch statement with example. 04
- (c) Write a recursive program to find factorial of a given number. 07
- OR**
- Q.4** (a) Differentiate function and macro. 03
- (b) Explain while loop and do-while loop with example. 04
- (c) Explain one dimensional array, two dimensional array and multi-dimensional array with their syntax and example. 07
- OR**
- Q.5** (a) Differentiate call by value and call by reference. 03
- (b) Explain structure within structure with example. 04
- (c) Explain error handling in file system with example. 07
- OR**
- Q.5** (a) Define Union in 'C' with example. 03
- (b) Define dynamic memory allocation. Explain malloc() and calloc(). 04
- (c) Explain the following File Handling functions: 1. fseek( ) 2. ftell( ) 3. fread( ) 4. fwrite( ) 5. fscanf( ) 6. fprintf( ) 7. rewind( ) 07

\*\*\*\*\*

**GUJARAT TECHNOLOGICAL UNIVERSITY****BE - SEMESTER-I & II(NEW) EXAMINATION – WINTER 2022****Subject Code:3110003****Date:07-03-2023****Subject Name:Programming for Problem Solving****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

|            |                                                                                                                                                                  | Marks     |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Q.1</b> | (a) List out and briefly explain basic building blocks of Computer system.                                                                                       | <b>03</b> |
|            | (b) Describe various symbols used for preparing flow chart.                                                                                                      | <b>04</b> |
|            | (c) List and explain different data types available in C.                                                                                                        | <b>07</b> |
| <b>Q.2</b> | (a) Demonstrate the working of bitwise shift and <i>sizeof</i> operator with an example.                                                                         | <b>03</b> |
|            | (b) Differentiate <i>while</i> and <i>do..while</i> loop.                                                                                                        | <b>04</b> |
|            | (c) Write a C program to check whether the entered character is alphabet, digit, space or any special character.                                                 | <b>07</b> |
|            | <b>OR</b>                                                                                                                                                        |           |
|            | (c) Write a C program to print following pattern:                                                                                                                | <b>07</b> |
|            | 1                                                                                                                                                                |           |
|            | 2 2                                                                                                                                                              |           |
|            | 3 3 3                                                                                                                                                            |           |
|            | 4 4 4 4                                                                                                                                                          |           |
| <b>Q.3</b> | (a) Find out error(s), if any in the following code and correct it:                                                                                              | <b>03</b> |
|            | <pre>int main() {     if(10%2==0){         break;     } }</pre>                                                                                                  |           |
|            | (b) Examine following code and give output of it.                                                                                                                | <b>04</b> |
|            | <pre>void main(){     int a=0;     while (a&lt;10) {         a++;         if (a%3==1) {             continue;         }         printf (" %d", a);     } }</pre> |           |
|            | (c) Write a C program to find factorial of a given number.                                                                                                       | <b>07</b> |
|            | <b>OR</b>                                                                                                                                                        |           |
| <b>Q.3</b> | (a) Find out error(s), if any in the following code and correct it:                                                                                              | <b>03</b> |
|            | <pre>int main() {     printf("%f %d", 7.0%5.0, 5    -2); }</pre>                                                                                                 |           |
|            | (b) Examine following code and give output of it.                                                                                                                | <b>04</b> |
|            | <pre>void main(){     int i=1;</pre>                                                                                                                             |           |

```

 for (i=10; i>=1 ; --i) {
 printf (“ %d”, i);
 i--;
 }
}

```

- (c) Write a C program to make sum of digits of a given number. (if input is 145, output should be 10) **07**
- Q.4** (a) Difference between *call by value* and *call by reference*. **03**  
 (b) Explain the concept of recursion with an example. **04**  
 (c) Write a C program to make sum of array elements. **07**
- OR**
- Q.4** (a) Briefly explain storage class *auto* and *extern*. **03**  
 (b) Explain *switch* statement with an example. **04**  
 (c) Write a C program to sort an array in ascending order. **07**
- Q.5** (a) Give the significance of *getc()*, *getw()*, *fscanf()*. **03**  
 (b) Compare structure and union. **04**  
 (c) Describe categories of User defined function. **07**
- OR**
- Q.5** (a) List down and briefly explain methods for dynamic memory allocation. **03**  
 (b) Write a C program to copy content of one file to other with the help of file handling functions. **04**  
 (c) Explain any four string handling functions with an example. **07**

**GUJARAT TECHNOLOGICAL UNIVERSITY****BE- SEMESTER-I & II(NEW) EXAMINATION – SUMMER 2023****Subject Code:3110003****Date:03-08-2023****Subject Name:Programming for Problem Solving****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

|            |                                                                                                                                                               | <b>Marks</b> |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Q.1</b> | (a) Design block diagram of computer and explain functionalities of various components of computer.                                                           | <b>03</b>    |
|            | (b) Write an algorithm to find maximum number for three given number.                                                                                         | <b>04</b>    |
|            | (c) Write a C program to compute Fahrenheit from centigrade ( $f=1.8*c+32$ ).                                                                                 | <b>07</b>    |
| <b>Q.2</b> | (a) Differentiate: Exit controlled Loop and Entry controlled Loop                                                                                             | <b>03</b>    |
|            | (b) Explain: <ol style="list-style-type: none"> <li>a) switch-case statement</li> <li>b) continue statement</li> </ol>                                        | <b>04</b>    |
|            | (c) Write a C program to find factorial of a number using recursion.                                                                                          | <b>07</b>    |
|            | <b>OR</b>                                                                                                                                                     |              |
|            | (c) Write a C program to generate first n number of Fibonacci series using recursion.( 0 1 1 2 3 5 8...)                                                      | <b>07</b>    |
| <b>Q.3</b> | (a) Explain else... if ladder with an example.                                                                                                                | <b>03</b>    |
|            | (b) List out all operators and explain any two in details.                                                                                                    | <b>04</b>    |
|            | (c) Write a function in the C program to return 1 if number is prime otherwise return 0.                                                                      | <b>07</b>    |
|            | <b>OR</b>                                                                                                                                                     |              |
| <b>Q.3</b> | (a) Explain Type casting in C with example.                                                                                                                   | <b>03</b>    |
|            | (b) Explain the structure of for loop with proper example.                                                                                                    | <b>04</b>    |
|            | (c) What is function? Differentiate: Call by value and Call by reference.                                                                                     | <b>07</b>    |
| <b>Q.4</b> | (a) Explain following terms: <ol style="list-style-type: none"> <li>a) break statement</li> <li>b) getch() function</li> <li>c) getchar() function</li> </ol> | <b>03</b>    |
|            | (b) Write a C program to print following patterns : <pre> * ** *** ****           </pre>                                                                      | <b>04</b>    |
|            | (c) Write a C program to copy one string to other without using string handling function.                                                                     | <b>07</b>    |

**OR**

- Q.4** (a) List any three header file with its usage. **03**  
(b) Write a program to print following patterns : **04**  
1  
22  
333  
4444  
(c) Define a structure type *struct* personal that would contain person name, date of joining and salary using this structure to read this information of 5 people and print the same on screen. **07**
- Q.5** (a) What is dynamic memory allocation? Show the use of `malloc()` and `calloc()` function with their syntax. **03**  
(b) What is structure? How it is different from union? Explain nested structure with example. **04**  
(c) Write a C program to check whether a number is prime or not. **07**
- OR**
- Q.5** (a) Explain following string handling functions: I. `strcat()` II. `strcmp()` III. `strlen` **03**  
(b) What is Pointer in C and how it initializes? State its advantages. **04**  
(c) Write a C program to check whether a three digits number is palindrome or not. **07**

\*\*\*\*\*

**GUJARAT TECHNOLOGICAL UNIVERSITY****BE - SEMESTER-I (NEW) EXAMINATION – WINTER 2023****Subject Code:3110003****Date:17-01-2024****Subject Name:Programming for Problem Solving****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

|            |                                                                                                                                                                                                                                                                                                                                                  | Marks     |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Q.1</b> | (a) Write an algorithm to check whether the entered number is Even or Odd.                                                                                                                                                                                                                                                                       | <b>03</b> |
|            | (b) Draw the flowchart to find the factorial of a number given by user.                                                                                                                                                                                                                                                                          | <b>04</b> |
|            | (c) Briefly explain different components of Computer system.                                                                                                                                                                                                                                                                                     | <b>07</b> |
| <b>Q.2</b> | (a) Give the output of following C code.                                                                                                                                                                                                                                                                                                         | <b>03</b> |
|            | <pre>int main(){     printf("%d", 15&lt;2);     printf("%d", 15&amp;&amp;2);     printf("%d", 7%10);     return 0; }</pre>                                                                                                                                                                                                                       |           |
|            | (b) Demonstrate the use of bitwise operators with an example.                                                                                                                                                                                                                                                                                    | <b>04</b> |
|            | (c) Explain C tokens in detail.                                                                                                                                                                                                                                                                                                                  | <b>07</b> |
|            | <b>OR</b>                                                                                                                                                                                                                                                                                                                                        |           |
|            | (c) Briefly explain different storage classes used in C with appropriate example.                                                                                                                                                                                                                                                                | <b>07</b> |
| <b>Q.3</b> | (a) Demonstrate the use of <i>ternary</i> operator with an example.                                                                                                                                                                                                                                                                              | <b>03</b> |
|            | (b) Give the output of following C codes.                                                                                                                                                                                                                                                                                                        | <b>04</b> |
|            | <div style="display: flex; justify-content: space-between;"> <div> <pre>1. int main(){     int i=10;     while(i&lt;10){         printf("%d", i);         i++;     }     return 0; }</pre> </div> <div> <pre>2. int main(){     int i=10;     do{         printf("%d", i);         i++;     }while(i&lt;10);     return 0; }</pre> </div> </div> |           |
|            | (c) Write a C program to print following pattern using loop.                                                                                                                                                                                                                                                                                     | <b>07</b> |
|            | <pre>5 4 4 3 3 3 2 2 2 2 1 1 1 1 1</pre>                                                                                                                                                                                                                                                                                                         |           |
|            | <b>OR</b>                                                                                                                                                                                                                                                                                                                                        |           |
| <b>Q.3</b> | (a) Differentiate between <i>break</i> and <i>continue</i> .                                                                                                                                                                                                                                                                                     | <b>03</b> |
|            | (b) Demonstrate the use of forward jump and backward jump with an example.                                                                                                                                                                                                                                                                       | <b>04</b> |
|            | (c) Explain <i>else if</i> ladder with an example.                                                                                                                                                                                                                                                                                               | <b>07</b> |

- Q.4** (a) Give the significance of *puts()*, *getchar()*, *getch()*. **03**  
 (b) Differentiate between *call by value* and *call by reference*. **04**  
 (c) Write a C program to check whether two strings are same or not. **07**

**OR**

- Q.4** (a) Give the output of following C code. **03**  

```
int main(){
 int val=20, *p;
 p = &val;
 printf("%d %d %d", val, *p, sizeof(p));
 return 0;
}
```

 (b) Demonstrate the use of recursion with an example. **04**  
 (c) Write a C program to find sum of digits for a given number using the concept of User Defined Function (UDF). (Hint: For number 3278, sum of digits is 3+2+7+8 = 20) **07**

- Q.5** (a) Differentiate between *structure* and *union*. **03**  
 (b) Briefly explain any two file handling functions with an example. **04**  
 (c) Write a C program to find maximum and minimum from an array of 10 elements. **07**

**OR**

- Q.5** (a) Give the output of following C code. **03**  

```
int main(){
 int arr[3][2]={1,2,3,4,5,6};
 printf("%d %d %d", arr[0][1], arr[1][0], arr[2][1]);
 return 0;
}
```

 (b) Briefly explain memory management functions. **04**  
 (c) Write a C program to copy one file to other. **07**

\*\*\*\*\*

**GUJARAT TECHNOLOGICAL UNIVERSITY**

**BE - SEMESTER-I & II (NEW) EXAMINATION – SUMMER 2024**

**Subject Code:3110003**

**Date:12-07-2024**

**Subject Name:Programming for Problem Solving**

**Time:02:30 PM TO 05:00 PM**

**Total Marks:70**

**Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

**MARKS**

- Q.1**
- |     |                                                                              |           |
|-----|------------------------------------------------------------------------------|-----------|
| (a) | Draw neat Block Diagram illustrating the Anatomy of a Computer System.       | <b>03</b> |
| (b) | Illustrate an Algorithm to check whether the given year is leap year or not. | <b>04</b> |

- |     |                                                                                                                  |           |
|-----|------------------------------------------------------------------------------------------------------------------|-----------|
| (c) | Define Flow chart, List the rules for designing a Flow chart and Rephrase the merits and demerits of flow chart. | <b>07</b> |
|-----|------------------------------------------------------------------------------------------------------------------|-----------|

- Q.2**
- |     |                                                                                                             |           |
|-----|-------------------------------------------------------------------------------------------------------------|-----------|
| (a) | List the rules to assign a name to an Identifier (Variable)                                                 | <b>03</b> |
| (b) | Illustrate the basic structure of a C Programs and highlight usual sections of <i>main</i> function.        | <b>04</b> |
| (c) | Construct a C program which receives number of days as input and converts them in to Year, Months and Days, | <b>07</b> |

**OR**

- |     |                                                                                          |           |
|-----|------------------------------------------------------------------------------------------|-----------|
| (c) | Construct a C program to swap the values of two integers with use of only two variables. | <b>07</b> |
|-----|------------------------------------------------------------------------------------------|-----------|

- Q.3**
- |     |                                                                                  |           |
|-----|----------------------------------------------------------------------------------|-----------|
| (a) | Contrast the Entry Control loop and Exit Control loop.                           | <b>03</b> |
| (b) | Construct a C program to find the largest number amongst three integers entered. | <b>04</b> |
| (c) | Construct a C program to check if entered integer is a prime number or note.     | <b>07</b> |

**OR**

- Q.3**
- |     |                                                         |           |
|-----|---------------------------------------------------------|-----------|
| (a) | Debug the following codes and show the output for this: | <b>03</b> |
|-----|---------------------------------------------------------|-----------|

```
For (i=0 ; i<=50 ; i++);
{
 printf("\n %d" , i)
}
```

*\*consider header files and declaration of variable is done.*

- |     |                                                         |           |
|-----|---------------------------------------------------------|-----------|
| (b) | Explain nested if-else statement with suitable example. | <b>04</b> |
| (c) | Construct a C program to print following pattern        | <b>07</b> |

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15.....
```

- Q.4** (a) List the types of Array. **03**  
 (b) Summarize the methods for initialization of One-Dimensional array. **04**  
 (c) Construct a C program to multiply 3X3 matrix. **07**
- OR**
- Q.4** (a) List the application of *gets*, *getchar* and *puts* functions. **03**  
 (b) Summarize the method for Declaration, initialization and Printing on screen of a String. **04**  
 (c) Define a Structures which contains details of a Cricketer: **07**
- Name of Player:
  - Team name:
  - Total run scored:
  - Batting average:
- Q.5** (a) Write True / False against following statements: **03**
- 1) Pointer is a variable which holds address of location where value of other variable is stored.
  - 2) Pointer holds the address in form of float or integer.
  - 3) The address of a variable can be stored in two or more pointers.
- (b) List the categories of User Defined Functions and Discuss any one of these. **04**  
 (c) Construct a C program which used a user define function that receives an integer and returns square and cube of it and prints in main function. **07**
- OR**
- Q.5** (a) What are the arithmetic operators that are permitted on pointers? **03**  
 (b) Show the use of *Malloc*, *Calloc*, *Free* and *Realloc* in context of Dynamic memory allocation. **04**  
 (c) List and discuss file handling functions used in C. **07**

\*\*\*\*\*

# GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II EXAMINATION – WINTER 2024

**Subject Code:3110003**

**Date:07-01-2025**

**Subject Name:Programming for Problem Solving**

**Time:10:30 AM TO 01:00 PM**

**Total Marks:70**

**Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q.1** (a) What is Software? List various types of Software. **03**
- (b) Define following terms: **04**
1. Compiler
  2. Runtime Error
  3. Dynamic Memory Allocation
  4. Recursion
- (c) Write an algorithm and draw the flowchart to calculate sum of digits of a given number N. **07**
- Q.2** (a) Explain the basic structure of a C program with an example. **03**
- (b) Evaluate given C programming Expressions : **04**
1. Ans =  $10 + 4 * 3 / 2$ ;
  2. Ans = while(1);
  3. Ans =  $3 * 2 > 3 + 1$ ;
  4. Ans =  $(2 + 4) > 3 \&\& 2 < 4$ ;
- (c) Write a C program to print the following pattern as output: **07**
- ```
A
B C
D E F
G H I J
```

OR

- (c) Write a C program that takes three coefficients (a, b and c) of a quadratic equation ; $(a.x^2 + b.x + c)$ as input and compute all possible roots and print them with appropriate messages. **07**
- Q.3** (a) Differentiate while loop and do..while loop with example. **03**
- (b) What is type conversion? Explain two types of conversion with examples. **04**
- (c) Write a C program which takes marks (between 0 to 100) of a student for N subjects. It should print grade of a student based on Percentage of Student's marks . **07**
- "A" for $\geq 85\%$,
- "B" between $\geq 70\%$ and $< 85\%$,
- "C" between $\geq 50\%$ and $< 70\%$,
- "PASS" between $\geq 35\%$ and $< 50\%$,
- "FAIL", if any one subject marks are < 35 .

OR

- Q.3** (a) What is function? How Call by value and Call by reference passes the argument in function. **03**
- (b) What are the steps in writing a recursive function in a program? **04**
- (c) Write a program in C to calculate the power of any number using recursion. **07**
- Q.4** (a) What is string ? How to declare string in C? **03**
- (b) Explain significance of following string functions with example. **04**
1. strcat()
 2. strlen()
 3. strcpy()
 4. strcmp()
- (c) Write a C program ,that reads a statement from user and prints the frequency of each of the distinct character from a-z. **07**

OR

- Q.4** (a) Compare Array, Structure and Union. **03**
- (b) Define Union. Describe how to declare, initialize and access members of Union with a programming example. **04**
- (c) Write a C program which takes two NxN matrices and performs matrix multiplication. **07**
- Q.5** (a) What is a pointer? What are the advantages of using pointer? **03**
- (b) Explain need of Dynamic Memory allocation. Compare malloc(), calloc() and realloc() functions. **04**
- (c) Write a C program to find the sum and mean of all elements in an array using pointer. **07**

OR

- Q.5** (a) What is File Pointer? What is significance of File Pointer? **03**
- (b) Explain signification of following functions in file operations. **04**
1. fseek()
 2. ftell()
 3. open()
 4. fread()
- (c) Write a C program to read ID and marks of n number of students from user and store them in a file. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I&II EXAMINATION – SUMMER 2025****Subject Code:3110003****Date:27-06-2025****Subject Name:Programming for Problem Solving****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

- Q.1**
- | | | |
|-----|---|-----------|
| (a) | Write an algorithm to find factorial of given number by user input. | 03 |
| (b) | Draw a flowchart to check whether the entered number is prime or not. | 04 |
| (c) | List and explain different data types available in C. | 07 |

- Q.2**
- | | | |
|-----|--|-----------|
| (a) | Demonstrate the working of ternary operator and sizeof operator with an example. | 03 |
| (b) | Differentiate between while and do..while loop. | 04 |
| (c) | Write a C program to print following pattern:
For e.g. input = 5
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5 | 07 |

OR

- | | | |
|-----|--|-----------|
| (c) | Write a C program to print following pattern:
For e.g. input = 5
*
**

***** | 07 |
|-----|--|-----------|

- Q.3**
- | | | |
|-----|--|-----------|
| (a) | Explain type casting with example | 03 |
| (b) | Differentiate between break and continue. | 04 |
| (c) | Write a program to find out the maximum number from integer array of 5 elements. | 07 |

OR

- Q.3**
- | | | |
|-----|---|-----------|
| (a) | Explain else...if ladder with example. | 03 |
| (b) | Explain getch(), putch(), gets(), puts() . | 04 |
| (c) | What is function? Differentiate between call by value and call by reference with example. | 07 |

- Q.4**
- | | | |
|-----|---|-----------|
| (a) | Explain nested structure with example | 03 |
| (b) | Explain following string handling functions:
1). strcat()
2). strlen() | 04 |
| (c) | Write a C program to check whether given integer number is palindrome or not? | 07 |

OR

- Q.4**
- | | | |
|-----|--|-----------|
| (a) | State the use of Union in 'C' with example | 03 |
| (b) | Briefly explain memory management functions. | 04 |

- (c) Write a C program to count total number of vowel in given string. (for example input string ="hello" , output=2) **07**

- Q.5** (a) Demonstrate the use of void * with example **03**
(b) Demonstrate the use of fopen() and fprintf() function of file handling **04**
(c) Write a c program to count total number of characters, words and line for any text file using file handling function **07**

OR

- Q.5** (a) Demonstrate the pointer arithmetic with example. **03**
(b) show the use of fclose() and fscanf() function of file handling **04**
(c) Write a c program to copy content of one text file to another text file **07**

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II EXAMINATION – WINTER 2025

Subject Code:3110003

Date:08-01-2026

Subject Name:Programming for Problem Solving

Time:02:30 PM TO 05:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Write an algorithm to check whether the entered number is Even or Odd.	03
	(b) Define following terms: 1) Application Software 2) System Software 3) Algorithm 4) Flowchart	04
	(c) Explain different type of operators used in c language with their precedence and associativity.	07
Q.2	(a) Discuss the need of break and continue statements with example.	03
	(b) Differentiate while and do..while loop	04
	(c) Write a C program to check whether the entered character is alphabet, digit, space or any special character.	07
	OR	
	(c) Write a C program to print following pattern: Input : 4 Output: 1 2 2 3 3 3 4 4 4 4	07
Q.3	(a) Explain switch statement with an example.	03
	(b) Write a C program to find the largest number amongst three integers entered	04
	(c) Write a C program to find factorial of a given number	07
	OR	
Q.3	(a) Explain nested if-else statement with suitable example.	03
	(b) Find out error(s), if any in the following code and correct it: int main() { if(25%2==0) { break; } }	04
	(c) Write a C program to make sum of digits of a given number. (if input is 145, output should be 10)	07

- Q.4** (a) List the types of Array **03**
(b) Summarize the methods for initialization of One-Dimensional array. **04**
(c) Write a C program to sort an array in ascending order. **07**

OR

- Q.4** (a) Demonstrate declaration and initialization of two dimensional array with suitable example. **03**
(b) Demonstrate the use of recursion with an example **04**
(c) Write a program in c for multiply two matrices A and B of dimensions M x N and N x O respectively and store the result in third matrix C. **07**

- Q.5** (a) Define dynamic memory allocation. Explain malloc() and calloc() with example. **03**
(b) What is pointer? Explain how pointers are declared and initialized. . **04**
(c) Write a c program to copy content of one file into another file. **07**

OR

- Q.5** (a) Differentiate call by value and call by reference. **03**
(b) Discuss Nested structure with example. **04**
(c) Write a c program to read a file and count total no of lines, total no. of words and total number of characters. **07**
